Bachelor Project Applied Computer Science
(Academic year 2025—2026)

Federated Learning

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Contents

1 Scope & Vision

1.1 Background

The project was carried out at company Result d.o.o. It is a software development company, but also focus on data, artificial intelligence, and business solutions. They work on many projects, some of which are European. One such is eEarly — an open platform for analysis and gathering of health data [1]. It was also the umbrella for this bachelor work.

European Union has many regulations and laws related to data, and there are even more for each individual country, organisation,... This introduces challenges for projects like eEarly that utilize artificial intelligence and deal with sensitive medical data. The motivation for my bachelor project is to find the bridge between sensitive data, artificial intelligence, and a complex regulatory landscape.

The direction of the artificial intelligence model was disease prediction. The main sources of data were APNEA HRV+SPO2 DATASET [2] and a smart medical ring Wellue O2Ring [1] that a person can wear while it continuously performs medical measurements. The artificial intelligence model used heart rate and oxygen saturation (SpO2) to predict whether a person has sleep-related breathing disorder called obstructive sleep apnea (OBS) [3].

1.2 Problem statement

In an ideal situation, an artificial intelligence model could train on all data without restrictions. In reality, besides strict European Union laws, such as General Data Protection Regulation (GDPR) [4], [5], each hospital normally keeps its data isolated. This reveals the main issue: An artificial intelligence model can — without considerable legal and data privacy effort — only train on one hospital. Consequences of this are poor model performance with less accurate predictions, and the new technologies are not being utilized fully [6]. From this follows the proposal, which is not to loosen the law, but to explore techniques like federated learning [7] to satisfy both the regulations and the artificial intelligence's demand for data.

1.3 Intention

The main intention of the project is to develop a federated learning proof-of-concept (POC) for the company, which can later be used as a starting point for other projects and might be fully implemented, possibly in a healthcare environment. It might also become a new privacy-oriented artificial intelligence solution that the company could offer to clients for commercialization as a reusable asset.

1.4 Scope within the bachelor project (MVP – need to have)

The scope can be seen on the Table 1 on page 3. The requirement descriptions include terminology introduced in Section 3. "Must have" features describe the minimal viable product (MVP); without these, the project is not deemed finished, unless blockers are clearly stated.

1.5 Ultimate scope (want to have)

Ultimate scope of the project is defined with features of "Should have" and "Could have" importance in the Table 1 on page 3. One can see that as importance lowers, features become closer to those of realistic setting in which federated learning would have been deployed, and more unknown challenges might appear.

[1] https://getwellue.com/pages/o2ring-oxygen-monitor

2 Software Requirements Specification

2.1 Target group

The primary target group of the project is developers. Most important aspect to consider for this target group is the knowledge to read code and its documentation. Their needs are not only to be able to understand code and its documentation, but also know how to interpret the results at the most basic level, and most important, they need to know how to run the project.

For the umbrella project eEarly, there are more potential stakeholders such as patients, whose sensitive data is guaranteed to remain private during the artificial intelligence training process, and doctors that would receive improved disease predictions. Perhaps there are even stakeholders from the legal perspective since the sensitive data is not moved and there is less possibility to breach current legislations and less need to adapt it. Though that is out-of-scope for the federated learning project.

2.2 Method

In order to discover, track, and prioritize all requirements, I was, especially in the beginning, asking a variety of questions to my mentor at Result d.o.o. and kept sorting them by importance using the MoSCoW framework [8]. Besides being very simple and clear, this method has proven itself in my previous projects, which is the main reason why I chose it as my main tool at this stage. All of this was done on Jira in a kanban board (see Appendix A.1), so that it has always remained organized and visible to me and mentors.

2.3 Requirements

Functional requirements can be seen in the Table 1. "Must have" requirements form the minimal viable product (MVP). Out of all requirements, I have created epics that can be seen in the list below. I have not used user stories, since the requirements are very broad with many tasks.

1. As a developer,
   I want to run a single-machine and non-containerized simulation of apnea predictions using federated learning,
   so that the whole structure is as simple as possible.

2. As a developer,
   I want to run a single-machine and containerized (with Docker) simulation of apnea predictions using federated learning,
   so that the whole project can be easily started by anyone on any machine.

3. As a developer,
   I want to run a multi-machine and containerized (with Docker) simulation of apnea predictions using federated learning,
   so that the structure comes closer to real and production-ready implementation.

4. As a developer,
   I want to run a multi-machine and containerized (on Kubernetes) simulation of apnea predictions using federated learning,
   so that the structure comes closer to real and production-ready implementation.

5. As a developer,
   I want to have a differential privacy mechanism implemented on top of federated structure,
   so that I can assure the system also secures weights and parameters that are sent to a central server.

Non-functional requirements:

- A developer needs to understand
  - the code,
  - the documentation,
  - how to interpret results at the most basic level, and
  - how to run the project.

Table 1: Functional requirements with their importance, higher rows correspond to higher priority.

Importance	**Requirement**
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.
Should have	Single-machine and containerized (with Docker) simulation of apnea predictions using federated learning.
Could have	Differential privacy mechanism implemented.
Could have	Multi-machine and containerized (with Docker) simulation of apnea predictions using federated learning.
Could have	Multi-machine and containerized (on Kubernetes) simulation of apnea predictions using federated learning.
Will not have	Non-basic visualizations.
Will not have	Advanced artificial model.
Will not have	Other security features like secure multiparty computation.
Will not have	Production ready features, e.g. blockchain technologies like swarm learning, compatibility with NVIDIA FLARE.

3 Preliminary Research

3.1 Federated Learning

Imagine there are researchers working on a project and they do not tell each other anything about themselves, their names, education, age... They only contribute with their current knowledge without revealing their personal information. This is what federated learning tries to achieve [9]. Later on I talk about global and local models, in this example, researchers play a role of local models, and the project would be a global model.

Federated learning is a *decentralized learning* technique, which means the learning is not accomplished centrally on a single device — normally a server — but is kept on the edge, close to the main source of the data. Since it is decentralized, this also leads to great benefits regarding *privacy* and regulations as data is not moved out of its safe zone. In practice, we accomplish this using two types artificial intelligence models: global and local. The algorithm [10] contains the following steps:

1. Global model is initialized.
2. Global model sends a copy of himself to all the local models. In a medical example, local models would be hospitals.
3. Local models train on sensitive (medical) data. There is no data exchanged between *any* of the models.
4. Local models send the weights to the global model. One can image they share their knowledge, without data.
5. Global model combines the weights — usually by averaging them —, and becomes better at predicting diseases, for example.
6. The process repeats from the second step. Each local model now becomes better as it receives the average "knowledge" from all others.

3.1.1 Types

There are two main types of federated learning that can be implemented based on the similarities between datasets of clients. A definition of a dataset is essential: A *dataset* is a collection of data, commonly structured into a format comparable to a spreadsheet where each rows represent a record (also called example). A record consists of values that represent either a feature or a label. A feature is input for a machine learning model, such as heart rate. A label is either output of a machine learning model — in a disease prediction examples would be 0 ("Obstructive sleep apnea not detected") or 1 ("Obstructive sleep apnea detected") — or an answer for a machine learning model while it is being trained on so called labeled data [11].

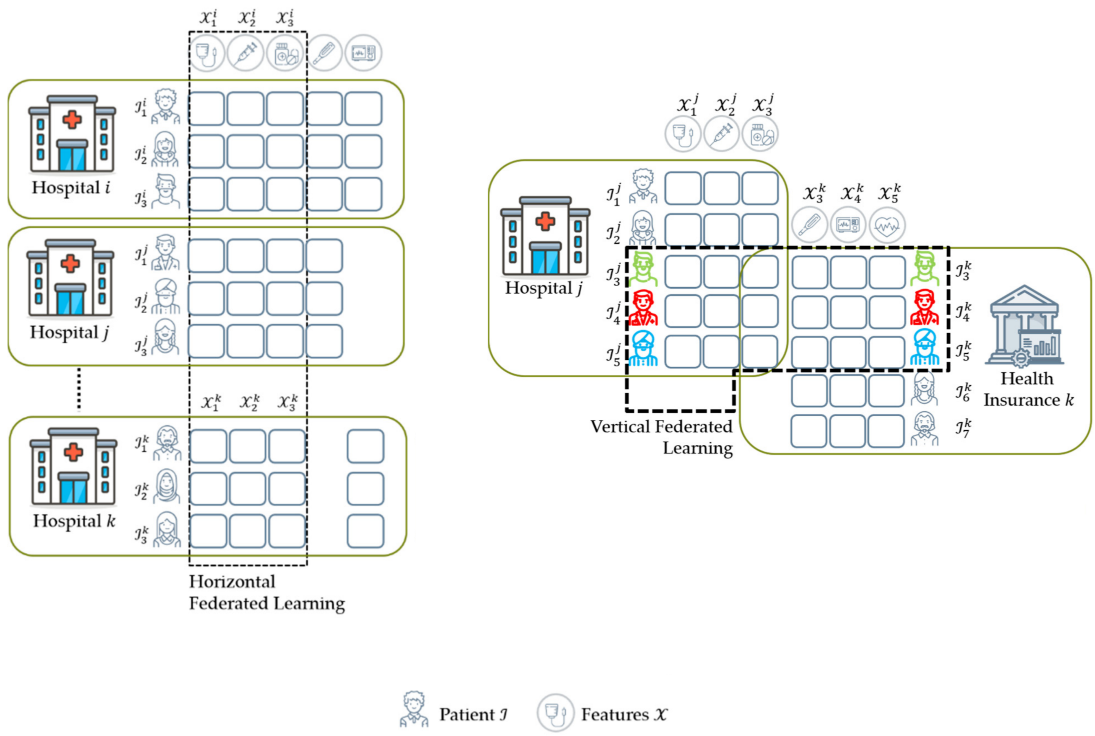
Horizontal Federated Learning In horizontal federated learning, clients' datasets describe same features meanwhile the data is different [5], [12]–[14]. For distinction with vertical federated learning, it is important here to keep in mind an almost obvious fact — By saying "the data is different" between clients, we mean that the feature that uniquely identifies a record, such as an email or a national identification number, is never duplicated across clients' datasets. For example, there are most likely many hospitals worldwide that have treated sleep apnea and have targeted same medical features, such as heart rate and oxygen saturation, but collected different data for different patients (see Figure 1). A federated learning model could predict sleep apnea based on all hospitals' client models.

Vertical Federated Learning In vertical federated learning, clients' datasets describe different data and different features, except for identification features such as identification number [5], [12]–[14]. For example, a country has a hospital for sleep disorders that collects features like patient national identification number, heart rate and oxygen saturation, and a cancer hospital that collects patient national identification number, x-ray scan images, blood information (visualized in Figure 1, although a health insurance is used instead of a cancer hospital) [15]. A federated learning model could connect the knowledge of both hospitals using patient identifier, find patterns, and become more holistic in its predictions.

Federated Transfer Learning In federated transfer learning, clients' datasets describe different data and different features, including the identification features [5], [12]–[14]. For example, two countries have two hospitals and there is very little overlap between features or data — the patients' identification is also mostly different — yet federated transfer learning can be applied in this scenario to combine their knowledge.

We will focus on horizontal federated learning prototype, because our features remain same across clients, namely hospitals.

Figure 1: Visualization of main types of federated learning in healthcare scenarios, sourced from [5]. On the left horizontal federated learning from different data with same features. On the right vertical federated learning from different data and mostly different features.



3.2 Related projects

Numerous similar projects with alike problems can be found, though there are some more noticeable and important.

In 2017, Google introduced federated learning while implementing the "Next word suggestion" feature on their Gboard. They took user interaction as their input and trained an artificial model to predict users next word. Without this innovating project, all the private typing data from millions of devices would have to be transferred to a central server. At this point, we can easily understand Google's main concerns of standard data pipeline: Bandwidth usage and latency [9], [16].

During COVID-19 pandemic, researchers all over the world have established connections and collaborated to create an artificial intelligence platform that predicts whether you have the disease based on CT (X-ray) chest scans. This was an initiative called Unified CT-COVID AI Diagnostic Initiative (UCADI), and the reason I am mentioning it here is that they have made it all possible by using federated learning [16], [17].

3.3 Related solutions

Main characteristics of federative learning described in Section 3.1 are decentralized learning and privacy; I use this two for finding related solutions below.

3.3.1 Decentralized Learning Techniques

Fog computing is an improved approach on top of federated learning. It originates from the network challenges of federated learning, focuses on layered architecture of communication between local and global model, and aims to reduce network traffic [18].

Being a decentralized technique, federated learning still has a central, global model that aggregates all parameter updates. *Gossip learning* — also called distributed federated learning (DFL) [19] — is even simpler in this sense, as it does not have a global model. The updates and aggregations are all taken care by the local models themselves [20].

Similar to gossip learning is *swarm learning*. The main difference is its use of blockchain peer-to-peer communication that is configured between the local models. This enables swarm learning to be a very related solution, with even better potential privacy and fault tolerance (because there is no single point of failure like the global model server) [19].

Centralized learning does not come into considerations because the movement of data to a central server is restricted by data regulations. Federated learning is perfect for the MVP, as it is simple and the core concept for the decentralization alternatives above that introduce unnecessary complexity for the project at this stage, although techniques like swarm learning would be necessary on a production-ready.

3.3.2 Privacy Preserving Techniques

In general, *homomorphic encryption* creates a possibility to make operations on encrypted data, without ever decrypting it. This can be applied both to normal and federated machine learning, although it is rather an addon than a replacement for the latter. The challenges include computational overhead and lack of support for some operations, thus finding a balance between privacy and performance is an important question [21], [22].

Secure multiparty computation allows to split private data from many participants, share it in this split form, and let any participant compute (public) functions, such as average and maximum without knowing any other participants' private data [23]. In federated learning, sharing weights with the central server still brings up some privacy concerns that can be handled by applying secure multiparty computation [24]. This adds some computation cost, along with even more communication overhead [21].

Differential privacy introduces random noise to private data while still maintaining its statistical properties [25]. This means that individual values like age are uninterpretable for any middleman between global and local models, thus guarding the system from inference and poisoning attacks. For example, inference attack would mean that a middleman can infer whether a person has a certain disease from the outputs of the global model [26]; data poisoning attack would mean that a middleman intercepts and modifies the weights sent to the global or local model, making the predictions of the model less accurate or corrupted [27].

All privacy alternatives mentioned above are strongly related, but optional for the scope of this work. They clearly point out main weak parts of the federated learning which have to be kept in mind in phases later on.

3.4 Required technology

There were minimal hardware requirements to develop the most important parts of the project. Must have requirements were satisfied only by using a single computer, and a dedicated remote server with a strong computing capabilities, specifically meant for faster model training, although it was not a necessity. Later on, another computer was added to configure the federated network in a multi-machine environment, making it more realistic. Ring sensor was not used, because other requirements were prioritized.

Software requirements consisted mostly of a programming language, integrated development environment and a virtualization technology. Since the apnea prediction model was already coded in a programming language Python (with Tensorflow machine learning library), I made sure this was also the case for the federated learning network. Federated Flower network was coded in Python and specifically configured for use with Tensorflow. Coding itself was done in IntelliJ and Visual Studio Code; the latter was used for easier SSH access to the remote machine learning server mentioned in the previous paragraph. Virtualization technology used after developing the minimal viable product was Docker, which made it possible to run all necessary code on separate computers without any other software. No other programming language or software was used or tested during this project.

3.5 Comparative study

From [28]–[30], I have extracted four main open-source frameworks suitable for the project. One of them, namely FedML, had vastly different opinions among papers, so I did not take this framework into consideration.

Flower: It's main characteristic is easy integration with different deep learning frameworks, and that it offers a beginner- and prototype-friendly entrance into federated learning [29], [31]–[33]. As Flower was new in 2025, research papers from that year showed it has lacked community support and documentation [29], [31], which has changed the next year, and is now apparently considered to have strong documentation [32]. It is especially well-suited for edge IoT devices and production ready [31], although additional privacy features have to be engineered manually [32], unlike with frameworks like NVIDIA FLARE.

NVIDIA Federated Learning Application Runtime Environment (NVIDIA FLARE): It's main characteristic is enterprise-readiness [31]–[34], with many built-in features, especially for privacy [34]. It is overall very good in all aspects (also documentation) [28], [31], which is the reason it might seem heavy for smaller projects [32]. The main disadvantages is that you are restricted to the NVIDIA ecosystem [31].

TensorFlow Federated (TFF): It's main characteristic is that is most widely used, but complex, only compatible with TensorFlow, and can be integrated into the least environments [29], [31].

The choice comes down to Flower or NVIDIA FLARE. Luckily, these two have recently announced collaboration and compatibility between each other [35]–[37], meaning Flower can be integrated into the NVIDIA runtime environment and immediately leveraging enterprise-ready environment from NVIDIA and simplicity from Flower. For this reason, Flower will be used to create the prototype.

3.6 Final selected technology / project tools

Project management tools used throughout this bachelor work where Jira, which I used for tracking work items and requirements, and Microsoft Excel, which I used for time tracking and diary of what I did each day. More details were described in Section 2.2.

Me and my coach respected agile principles, and the flow of the project followed scrumban development methodology — a combination of scrum and kanban. Since I was the only developer, most scrum events were skipped, although I did have daily meetings with my company coach. The way tickets were tracked is completely based on kanban, having a board with todo, doing, and done columns.

3.7 Choice reflection

Decisions described above were based on experience, research and communication with my company mentor. The technologies were correct, which was proven by the fact that we did not have to change any during the development. For me, it remains a question, whether these technologies could withstand the enterprise needs, and what weak points would be revealed during its complete deployment. The project tools were great, especially Jira that offer a powerful way to increase productivity and insight into the project organization and status, without unnecessary features.

4 Reflection

4.1 General Reflection

4.1.1 Achieved objectives

The project requirements written in Table 1 turned out to be well-thought as majority was implemented. By successfully establishing single-machine non-containerized simulation of apnea predictions minimal viable product was delivered. This was further more extended — using Docker — to multi-machine containerized setup. On top, additional security was configured with TLS certificate requirement between the all communicating nodes. Although this security might not be necessary since it was done in a single network, the federated setup is ready to be tested in a multi-network environment where this would become a must. All requirements with associated completion status can be seen in Table 2.

Table 2: Functional requirements with their importance and completion status, higher rows correspond to higher priority.

Importance	Requirement	Implemented
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.	Yes
Should have	Single-machine and containerized (with Docker) simulation of apnea predictions using federated learning.	Yes
Could have	Differential privacy mechanism implemented.	No
Could have	Multi-machine and containerized (with Docker) simulation of apnea predictions using federated learning.	Yes
Could have	Multi-machine and containerized (on Kubernetes) simulation of apnea predictions using federated learning.	No
Will not have	Non-basic visualizations.	No
Will not have	Advanced artificial model.	No
Will not have	Other security features like secure multiparty computation.	No
Will not have	Production ready features, e.g. blockchain technologies like swarm learning, compatibility with NVIDIA FLARE.	No

However, there are still some unresolved issues. One of them being the aggregated evaluation metrics showing presumably wrong results when it comes to precision and recall scores — these are always zero, in contrast to centralized training where this does not happen. On the other hand, the train metrics are correct, which is why the task is deemed achieved, besides that my focus was not on improving model performance. Perhaps testing the final model on all data would be an indicator about the direction of the issue — were the metrics not aggregated correctly, or is there an issue solely in the evaluation code.

A summary of final metrics from both centralized and federated approaches can be seen in Table 3. The differance between these two machine learning techniques, in terms of metrics, turn out to be very minor, around 0.03 ± 0.05, with federated learning mostly having slightly lower values; except for the test metrics, where we suspect a minor bug in the code. Results of another test can be found in Appendix B.

4.1.2 Test framework / technique

No automated testing was written during the project. The model is manually tested by using evaluation and prediction scripts, which isolate some patients measurements from the training procedure and later on execute predictions on them to give us representative results of model performance. The federated framework part of the project also does not have any special tests. Important to note is that the different hospitals, in different Docker containers do not share any patients measurements data during training and testing, which is also the goal of federated learning.

In production, besides automated testing of data pipeline, privacy that federated learning advertises should be thoroughly tested not to show any vulnerabilities of attacks described in Section 3.3.2, such as inference attack.

4.1.3 Security

The whole concept of federated learning is build for the purpose of securing sensitive data by not sharing it to third parties, which is also what has been implemented in this proof-of-concept. In

Table 3: Results of centralized and federated learning approach and their comparison. In a federated setting, the test consisted of 3 hospitals (clients), 5 rounds and 15 epochs. In a centralized setting, it consisted of 15*5*3 (255) epochs. Hyperparameters were the same for both approaches. Number 0 or 1 after metric name indicates whether it was calculated for patients with or without obstructive sleep apnea.

Metric	Centralized	Federated	Difference
train_accuracy	0.80	0.79	0.01
train_f1_1	0.52	0.51	0.01
train_f1_0	0.78	0.74	0.02
train_precision_1	0.43	0.40	0.03
train_precision_0	0.86	0.87	0.01
train_recall_1	0.64	0.70	0.06
train_recall_0	0.72	0.65	0.07
test_accuracy	0.76	0.83	0.07
test_precision	0.35	0.00	0.35
test_recall	0.48	0.00	0.48
avg_train_difference			0.03
avg_test_difference			0.3
avg_difference			0.111

a multi-machine setup, Docker containers to not share any private patients measurement data between each other. They merely exchange the knowledge in terms of model weights. Most additional security features mentioned in Section 3.3.2 were left out in favor of expanding the project towards the infrastructural direction by containerizing the implementation. While developing that, secure TLS communication was enforced in order for the gRPC communication between nodes to become encrypted and weights cannot be directly read by a middleman.

Possible improvements in the project's security mostly entail differential privacy, which is already supported by Flower out-of-the box and offers a mostly straight-forward implementation. More difficult security features to further research and develop would be secure multiparty computation and homomorphic encryption. In case of progressing the project to Kuberenetes, it is important to understand how it can negatively affect the security that Flower brings and use an appropriate framework such as kubeFlower [38].

4.1.4 Improvements

Starting with a non-functional improvement: readability. The model code I took was a few months old and there is a more readable and clean model that a colleague at the company has developed since. Plugging in that model would be a highly beneficial next step. Afterward, running the federated network using docker on *separate* networks would be one step closer to real implementation. Adding TLS certificates issued by an official authority like Let's Encrypt, instead of using self-signed certificates like I have done now is a mandatary move; Let's Encrypt also offers test certificates, which could be great while playing around with this part of security. Finally, reducing bandwidth and model weight size sent over network even more might be an interesting topic of research.

4.1.5 Added value of the project

The greatest added value for the project is definitely privacy. The fact of sending knowledge (model weights) instead of data is crucial for fewer conflicts with regulations. This brings more data for a better performing machine learning model and less effort and time required to align with the law.

4.1.6 Future opportunities and next steps

This project could be potentially adapted to any other machine learning project, especially those with privacy concerns or many IoT devices, sensors, moving devices, and so on. It could be used in many optimization scenarios, such as improving car safety features by learning on a fleet of vehicles without directly sending the data over the network, enhancing traffic light systems to reduce traffic jams, just to name a few possibilities. Mastering the implementation of a federated system might bring a company such as Result d.o.o a wide range of real-world project opportunities.

4.2 Personal Reflection

Graphics card Completely remove the Ring out of scope Nothing else much

Big easly understand that their data won't be seen by anyone except their hospital like it used to be less afraid of ai stealing data stationary

ecomony – bigger growth policy — less regulation burden, more compliant approach educatoin – not much impact, except perhaps special educational tools build on machine learning, e.g. Anki.

less bandwidth less resources/electricity perhaps this new approach improves predictions by tons of sensors, weather. better smart applianaces more efficient energy systems, like cars, or any process that could benifit from a federated network. process efficiency -¿ energy reduction

A lot, becasu not working with these systems before how little I know about the machine learning models in other words, how vast this field is. from that interest to learn, mostly the optimization challenges interest, that I always had.

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4.3 Feedback

glad that I had mix of software development during traineeship and ML in bachelor project glad manja gave me a change to learn hyperparameter tuning during trainsship for a short preiod to know the field better. Perfect because strong connection to such a good company.

Tell new students the reality of removing 2/3 of their expected outcome/requirements. I started worked n hours for it before the start, have achieved mvp and 1 should have. Not having strict deadlines for submitting on toledo is great. The way that the structure of the diploma document is, with all the preliminary research, it takes lot of time. Less for implementation. Research/development templates? Preliminary research takes most of time Introduce students to powerful citation tools like Zotero.

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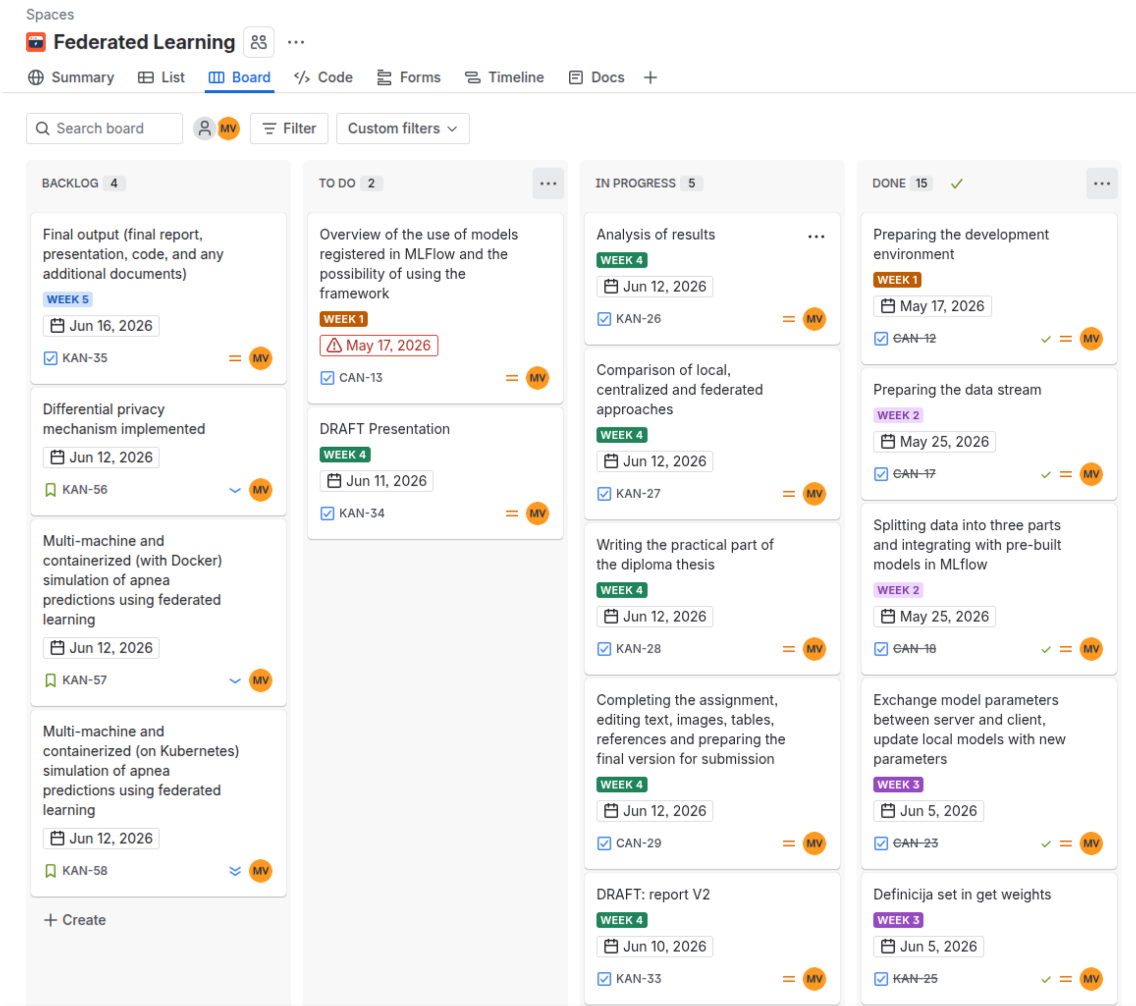
Appendices

A Story mapping

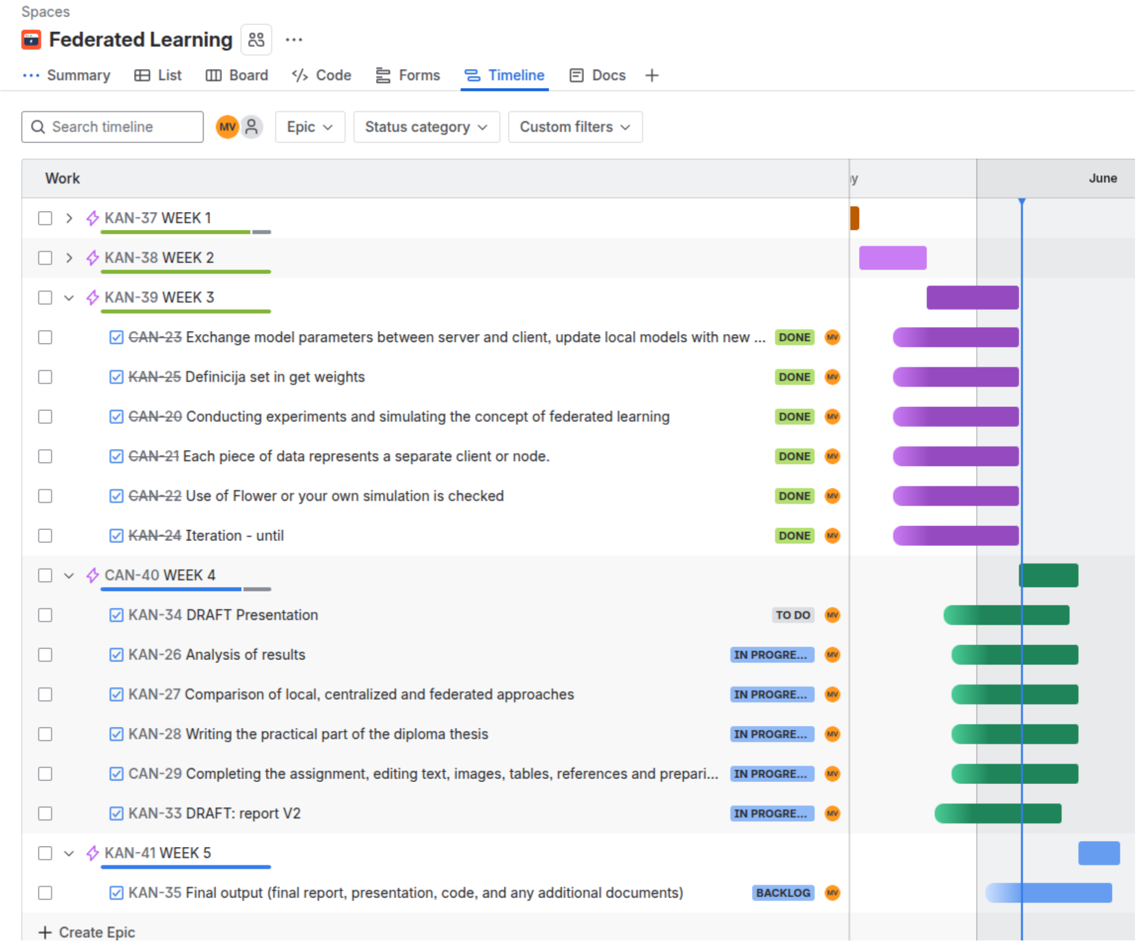
A.1 Kanban Board

Figure 2: Translated using Google Translate.



A.2 Timeline

Figure 3: Translated using Google Translate.



B Second Test

Table 4: Results of centralized and federated learning approach and their comparison. In a federated setting, the test consisted of 3 hospitals (clients), 5 rounds and 15 epochs. In a centralized setting, it consisted of 15*5*3 (255) epochs. Hyperparameters were the same for both approaches. Number 0 or 1 after metric name indicates whether it was calculated for patients with or without obstructive sleep apnea.

Metric	**Centralized**	**Federated**	**Difference**
train_accuracy	0.80	0.80	0.00
train_f1_1	0.49	0.52	0.03
train_f1_0	0.80	0.77	0.03
train_precision_1	0.44	0.42	0.02
train_precision_0	0.84	0.87	0.03
train_recall_1	0.57	0.68	0.11
train_recall_0	0.76	0.69	0.07
test_accuracy	0.79	0.83	0.04
test_precision	0.39	0.00	0.39
test_recall	0.41	0.00	0.41
avg_train_difference			0.041
avg_test_difference			0.280
avg_difference			0.113

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